

Deterministic Clinical Retrieval & Reasoning Architectures

Eliminating LLM-based inference from constraint-heavy clinical query resolution over large medical corpora

The Core Challenge

Three things make implicit clinical constraint resolution hard without probabilistic inference:

- 1 **Vocabulary gap, not just synonym gap.** “Aged 10–20” needs to resolve to multiple overlapping, non-nested clinical categories (pediatric $\approx$ 0–18, adolescent $\approx$ 10–19, puberty is physiologically variable and not a clean age band at all) — and the corpus may use any of them inconsistently. A literal age-range parser fails immediately; you need a controlled vocabulary layer that can hold “adolescent” and “10–20 years” as the same node with defensible, cited boundary logic.
 - 2 **Constraint composition, not constraint lookup.** Real clinical queries aren’t single facts — “Disease X, age 10–20” implicitly drags in weight-based dosing, pregnancy status exclusions, renal/hepatic contraindication checks, and drug-drug interaction hierarchies. This is a constraint satisfaction problem, not a search problem — you need something that can compose multiple simultaneous logical constraints and refuse to answer if any required constraint is unresolved, rather than silently guessing.
 - 3 **Traceability requirement.** For clinical safety you don’t just need an answer, you need a provable chain: which sentence, which page, which ontology mapping, which rule fired. That is the actual reason LLMs are a poor fit here — not that they are inherently “bad,” but that a probabilistic generation step breaks the audit chain a clinician or regulator would need to verify the answer independent of the model.
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1. Ontology-Anchored Description Logic Reasoner

High-Level Concept

Every clinical concept in the corpus (and in the query) is normalized to UMLS/SNOMED-CT concept IDs at ingestion time. Retrieval becomes a Description Logic subsumption query, not a text match.

How It Solves the Scenario

“Age 10–20” is parsed by a rule-based NER+N layer (not an LLM) into a normalized age range, which is mapped via a curated age-band ontology extension (SNOMED does not natively encode age ranges well, so this typically requires a custom OWL extension) to concept nodes like `Pediatric_Population`, `Adolescent`. A reasoner (HermiT/Pellet) computes subsumption: is `patient_age_10_20` a subclass of, or overlapping with, `Adolescent`? Text spans tagged with `Adolescent` or its ontological parents/children are retrieved. Drug contraindications are separate OWL axioms (`DrugX contraindicated_in Pediatric_Population`) — the reasoner checks these deterministically, not via inference over free text.

The Workflow

Query → rule-based entity/age extraction → concept normalization against UMLS/SNOMED → DL subsumption query against ontology → indexed text spans tied to matched concept nodes retrieved → contraindication axioms checked → structured answer with concept-ID provenance for every claim.

Clinical Rationale

Every inference step is a formally verifiable logical entailment — a reasoner either proves the subsumption relation or it doesn't. No probabilistic step exists between query and answer.

Key Trade-offs & Bottlenecks

The bottleneck is curation, not compute. SNOMED/UMLS don't cleanly encode fuzzy age bands or overlapping category boundaries (is a 19-year-old “pediatric”? depends on the guideline). Someone has to hand-author or validate these axioms, and DL reasoners scale poorly (often super-linear) as axiom count grows into the tens of thousands needed for a 10k-page corpus. Also brittle to corpus text that isn't already written in ontology-normalizable language — dense narrative prose (case reports, discussion sections) resists clean concept tagging.

2. Deterministic Graph-Relational Constraint Pipeline

High-Level Concept

Instead of reasoning over an ontology at query time, pre-compile the entire corpus into a property graph (Neo4j) where every clinically relevant fact is a typed edge, e.g. (DrugX)-[TREATS {age_min:10, age_max:19, evidence_span:"p.402¶3"}]->(DiseaseX). Query answering becomes a graph traversal + filter, not logical inference.

How It Solves the Scenario

At ingestion, a deterministic extraction pipeline (dependency parsing + rule-based relation extraction over UD parse trees, not an LLM) pulls out age/demographic-qualified clinical assertions and writes them as edges with explicit numeric ranges rather than category labels — sidestepping the “adolescent vs pediatric” ambiguity entirely by extracting the actual numbers the source text states, falling back to a curated age-band lookup table only when the source text is categorical (“in pediatric patients”) rather than numeric. Query time is a Cypher traversal filtered on the numeric range.

The Workflow

Query → parameter extraction (disease, numeric age range) → Cypher query construction (parameterized, not templated by an LLM) → graph traversal returns matching edges + source spans → contraindication edges checked via a second traversal → synthesis is literally the query result set, formatted, with every returned fact carrying its source span.

Clinical Rationale

No ontology-subsumption ambiguity — ranges are numeric and comparisons are simple arithmetic, about as verifiable as it gets. The traversal is inspectable and re-runnable by anyone with Cypher access.

Key Trade-offs & Bottlenecks

The extraction pipeline is the real bottleneck — rule-based relation extraction over free clinical prose has much lower recall than an LLM would (idiomatic phrasing, negation scope, hedged language like “may be considered in adolescents” all trip up hand-written extraction rules), so this architecture trades hallucination risk for silent omission risk: facts the extractor fails to parse simply never enter the graph, with no signal that something was missed. Also requires ongoing curation as new literature is ingested — extraction rules need continual expansion for new phrasing patterns.

3. Compiler-Style AST Query Parser + Constraint Solver

High-Level Concept

Treat the clinical query itself as a small formal language. Parse it into an abstract syntax tree the way a compiler parses source code, then evaluate that AST against a constraint solver (CSP or SMT, e.g. Z3) rather than free-text matching at any stage.

How It Solves the Scenario

“Treatment for Disease X in patients aged 10 to 20” is parsed by a grammar (context-free grammar over a controlled clinical query syntax, or a dependency-parse-to-AST compiler for near-natural-language input) into a typed tree: `Query(Intent=Treatment, Disease=X, Constraint=AgeRange(10,20))`. The AST is compiled into a set of first-order logic constraints. Corpus facts (extracted once, offline, into a fact database, similar to the graph approach but stored as logical predicates) are loaded into an SMT solver alongside the query constraints. The solver finds all facts satisfying the constraint set, including facts stated in overlapping categorical bands — `Adolescent ⊆ AgeRange(10,19)` is itself encoded as an axiom, so the solver, not a lookup table, derives the overlap.

The Workflow

Query → grammar-based parse into AST → AST compiled to first-order-logic constraint set → constraint set + corpus fact-base loaded into solver → solver returns satisfying assignments (which facts, which evidence spans) → unsatisfiable-core reporting if no facts satisfy the query (explicit “no answer found” rather than a forced guess) → structured synthesis from the satisfying set.

Clinical Rationale

SMT solvers produce not just an answer but a proof of satisfiability and, critically, can report unsatisfiability cleanly — the system can honestly say “no facts in this corpus satisfy this exact constraint combination” instead of returning the nearest approximate match, a genuinely distinct safety property from the other approaches.

Key Trade-offs & Bottlenecks

Grammar coverage is brittle — natural clinical queries have huge syntactic variety, and anything outside the grammar either fails to parse or requires an expanding, hand-maintained grammar (the same curation burden as approach 1, moved earlier in the pipeline). Solver performance can degrade sharply as the fact-base and constraint complexity grow, particularly with hierarchical/overlapping category axioms clinical medicine actually has (age bands, dosage-by-weight-by-renal-function chains).

4. Multi-Stage Deterministic Retrieval Cascade

High-Level Concept

Rather than one clever reasoning layer, chain several narrow, individually-verifiable deterministic stages, each with a hard fallback and explicit failure signaling — optimizing for the system knowing when it doesn't know, rather than for any single stage being maximally clever.

How It Solves the Scenario

Stage 1: rule-based NER+N tags all entities in the query against UMLS with confidence scores derived from string-match quality (exact match, synonym-table match, abbreviation-table match — each a distinct deterministic lookup, never inferred). Stage 2: a curated age-band crosswalk table (explicitly versioned and clinician-reviewed, e.g. “WHO adolescent definition: 10–19,” “AAP pediatric definition: 0–18,” both retrievable with source citation) maps the numeric range to all applicable categorical labels rather than picking one — deliberately over-including rather than resolving ambiguity silently. Stage 3: a boolean/BM25-style structured index (metadata-filtered, not vector-similarity-based) retrieves candidate passages tagged with any of the stage-2 labels. Stage 4: a small, hand-written rule engine (CLIPS/Drools) checks each candidate passage's associated structured contraindication/dosing facts against the full constraint set and either passes it through or rejects it with a logged reason.

The Workflow

Query → NER+N (confidence-scored) → age/demographic crosswalk (multi-label, cited) → structured boolean retrieval → rule-engine constraint verification per candidate → ranked output where every passage carries its full stage-by-stage provenance trail, and low-confidence stage-1 matches trigger an explicit “ambiguous term, human review recommended” flag instead of proceeding.

Clinical Rationale

No stage does “creative” reasoning — each is independently testable and each failure mode is visible rather than hidden inside a black box, which matters for regulatory audit (you can point to exactly which stage and which lookup table produced a given result).

Key Trade-offs & Bottlenecks

This is the most operationally realistic of the four but also the least “clever” — it doesn't actually solve the hard cases the other approaches attempt (novel constraint combinations, deep composition of multiple contraindications), it mostly punts them to explicit human review. Curation burden is spread across every stage rather than concentrated, which is easier to maintain incrementally but harder to reason about holistically, and the confidence-scoring thresholds in stage 1 are themselves a place where subtle miscalibration can cause silent under- or over-triggering of human review.

Cross-Cutting Observation

All four approaches share the same real bottleneck, just relocated: someone has to hand-curate the mapping between natural clinical language and the formal structures the system reasons over (ontology axioms, graph extraction rules, query grammars, or crosswalk tables). None of them eliminate that labor — they only differ in where it lives and how visible its failure modes are. This is worth stating plainly in any

writeup, since “fully deterministic” can read as “curation-free” to a non-technical stakeholder, and it isn’t.